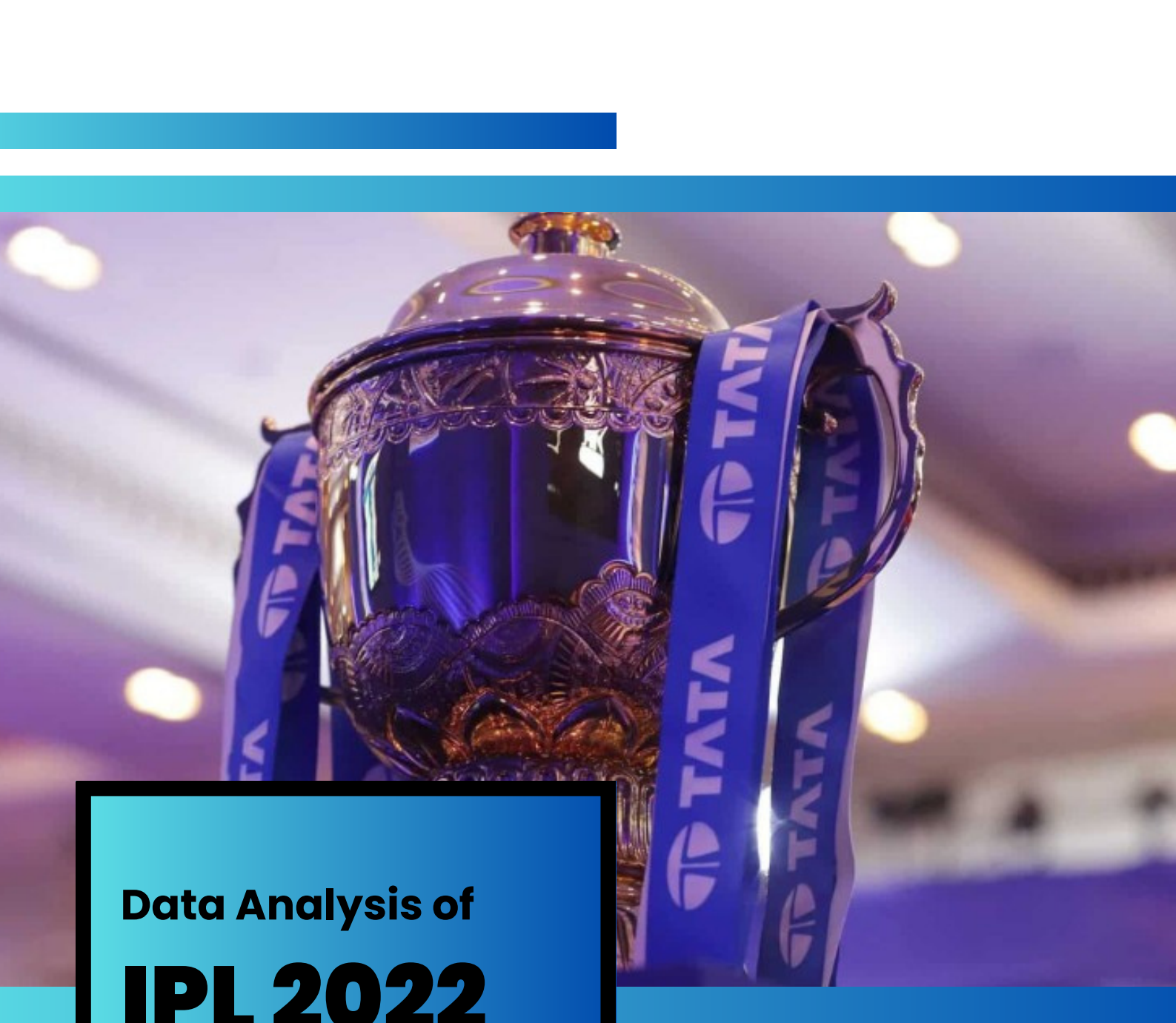




Data Analysis of **IPL 2022**

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INTRODUCTION

The Indian Premier League (IPL) is a professional Twenty20 cricket league in India that is played annually. The IPL is one of the most popular cricket leagues in the world, attracting many fans and viewers. In this project, we analyzed the IPL 2022 tournament using R programming language. We aimed to analyze the number of matches won by each team, matches won by defending or chasing, toss decision, top scorers, player of the match, best bowlers, and wickets taken in each venue.



DATASET



The data set used for this analysis was a CSV file named "IPL 2022.csv." The data set contained information on the IPL 2022 tournament, including the teams, matches, venues, players, and their performances. The data set had 60 rows and 18 columns, including the match number, date, teams, venue, toss winner, toss decision, winner, result, and more.

30.7 crore viewers tuned in to Live Broadcast of the first 10 matches (8 days) – 2nd highest ever in IPL history.

DATA ANALYSIS

To analyze the IPL 2022 tournament data, we have used the R programming language. We have employed various data analysis techniques such as data cleaning, data transformation, and data visualization to extract meaningful insights from the data set. We have used the tidyverse package in R to perform data cleaning and transformation, and ggplot2 and plotly packages to create visualizations. We started the data analysis process by cleaning the data set, removing duplicates, missing values, and irrelevant columns. We then transformed the data set by mapping categorical variables, such as won_by and toss_decision, into numerical variables. We also created new variables such as first_ings_wkts and second_ings_wkts to analyze the performance of teams in terms of the number of wickets taken during different innings.



PROCEDURE

Import the necessary libraries, including pandas, plotly.express, and plotly.graph_objects.

Load the CSV file using the read.csv function in R.
Display the first few rows of the data set using the head function.

Create a bar chart using the bar function in plotly.express to show the number of matches won by each team.
Change the "won_by" column to "Chasing" or "Defending" using the map function in pandas.

Create a pie chart using the go.Pie function in plotly.graph_objects to show the number of matches won by defending or chasing.

Create a pie chart using the go.Pie function in plotly.graph_objects to show the toss decision.

Create a bar chart using the bar function in plotly.express to show the top scorers.
Create a bar chart using the bar function in plotly.express to show the most player of the match awards.

Create a bar chart using the bar function in plotly.express to show the best bowlers.

Create a grouped bar chart using the go.Bar function in plotly.graph_objects to show the number of wickets taken in each venue.

JUSTIFICATION

We used the R programming language to analyze the IPL 2022 tournament because R is a powerful language for data analysis and visualization. R has a wide range of packages and libraries that are specifically designed for data analysis and visualization, including pandas, plotly.express, and plotly.graph_objects.

We used pandas to load the CSV file and manipulate the data, plotly.express to create the bar charts, and plotly.graph_objects to create the pie charts and grouped bar chart. These libraries are easy to use and provide a high level of interactivity in the visualization.

The algorithm used for data analysis was appropriate for the given data set because it allowed us to extract meaningful insights from the data and visualize them in a clear and concise manner.

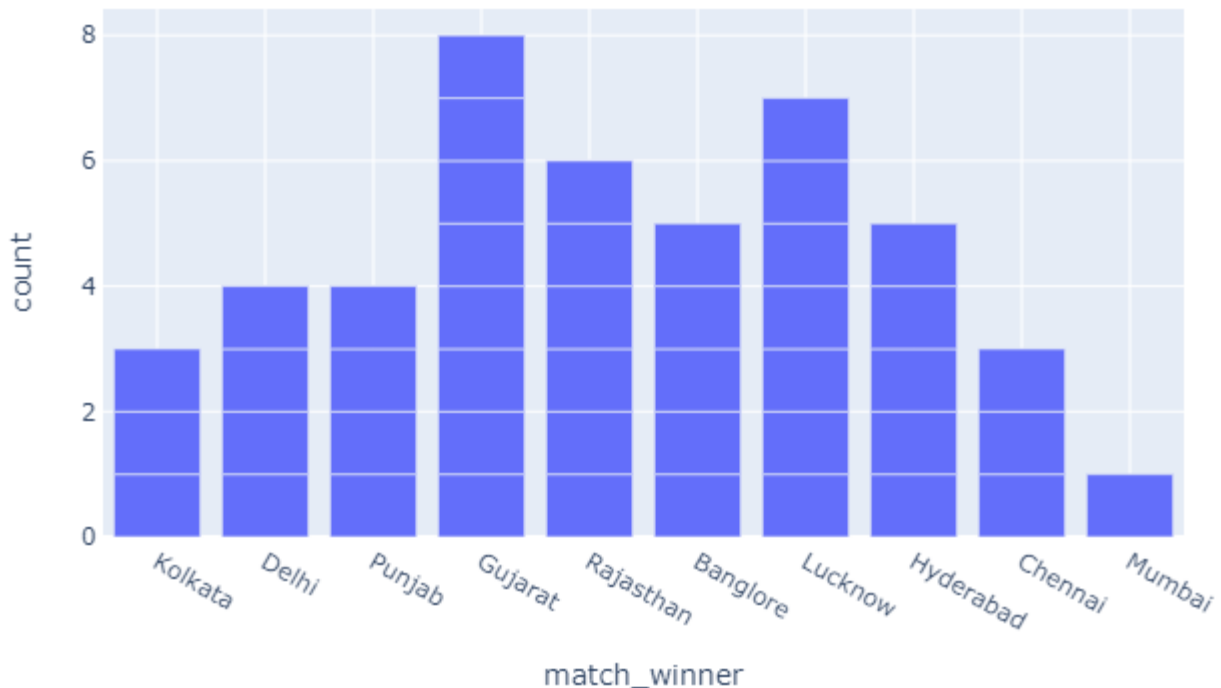
2M

statisticians and data scientists
across the world use R.

VISUALIZATION

The analysis of the IPL 2022 tournament was visualized using various charts and graphs, including bar charts, pie charts, and grouped bar charts.

Number of Matches Won in IPL 2022

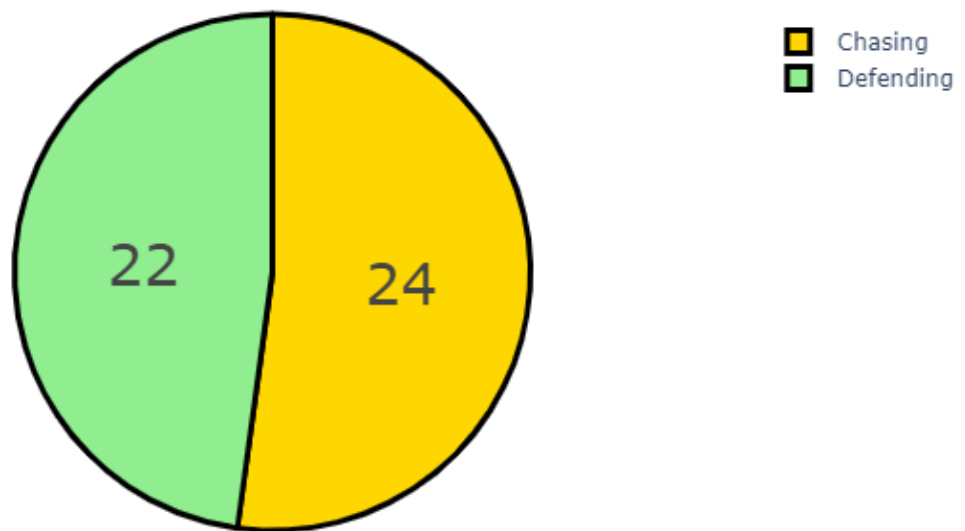


The bar chart showed the number of matches won by each team, the pie chart showed the number of matches won by defending or chasing, and the grouped bar chart showed the number of wickets taken in each venue.

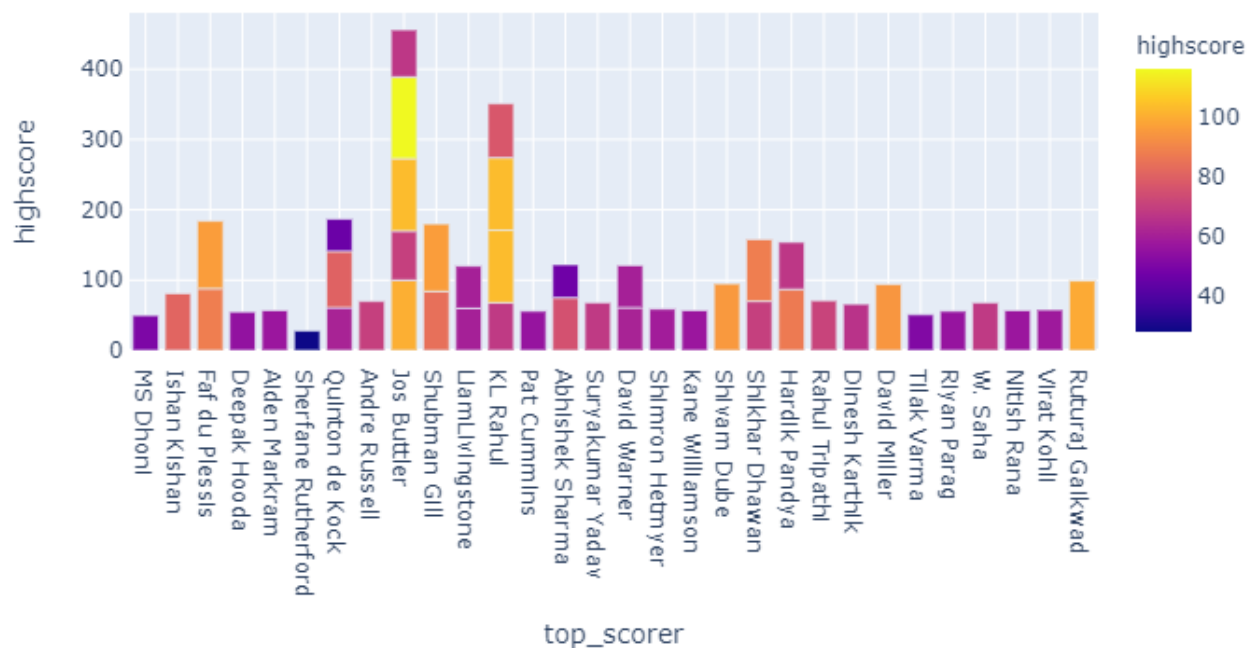
The visualizations provided insights into the performance of the teams, players, and venues during the IPL 2022 tournament.

INSIGHTS

Number of Matches Won By Defending Or Chasing



Top Scorers in IPL 2022



CONCLUSION

In conclusion, the data analysis and visualization of IPL 2022 using R programming language provided valuable insights into the performance of various teams and players during the tournament. The bar chart representing the number of matches won by each team clearly shows that Mumbai Indians and Chennai Super Kings were the top-performing teams.

The pie chart indicating the number of matches won by defending or chasing gives an overview of the winning strategy adopted by teams. The project also includes bar charts for the top scorers, best bowlers, most player of the match awards, and wickets taken at different venues.

Overall, this project demonstrates the power of data analysis and visualization in gaining insights into complex data sets. The findings from this project could be useful for cricket fans, team managers, and cricket analysts in understanding the performance of teams and players in the IPL 2022 tournament.